

VEDA TECHNOLOGY – DATA ANALYTICS INTERNSHIP

TASK 21 – BASIC SQL SELECT

Platform: Google Colab **Database:** SQLite **Table:** students

Task Objective: Practice SQL SELECT, WHERE, ORDER BY and basic filtering using a student dataset.

Student Dataset

Student_ID	Student_Name	Department	Year	Marks	City
101	Arun	AI & DS	1	87	Namakkal
102	Divya	CSE	2	92	Salem
103	Kavin	ECE	1	76	Erode
104	Harini	AI & DS	3	95	Karur
105	Priya	IT	2	81	Coimbatore
106	Rahul	CSE	3	89	Chennai
107	Keerthana	AI & DS	2	91	Erode
108	Vijay	ECE	4	73	Salem
109	Nandhini	IT	1	88	Namakkal
110	Sanjay	AI & DS	4	97	Karur
111	Meena	CSE	1	79	Erode
112	Ashwin	IT	3	84	Chennai
113	Swetha	ECE	2	90	Coimbatore
114	Dinesh	CSE	4	94	Salem
115	Pavithra	AI & DS	1	86	Namakkal

Query 1 - SELECT ALL RECORDS

```
SELECT * FROM students;
```

Student_ID	Student_Name	Department	Year	Marks	City
101	Arun	AI & DS	1	87	Namakkal
102	Divya	CSE	2	92	Salem
103	Kavin	ECE	1	76	Erode
104	Harini	AI & DS	3	95	Karur
105	Priya	IT	2	81	Coimbatore
106	Rahul	CSE	3	89	Chennai
107	Keerthana	AI & DS	2	91	Erode
108	Vijay	ECE	4	73	Salem
109	Nandhini	IT	1	88	Namakkal
110	Sanjay	AI & DS	4	97	Karur
111	Meena	CSE	1	79	Erode
112	Ashwin	IT	3	84	Chennai
113	Swetha	ECE	2	90	Coimbatore
114	Dinesh	CSE	4	94	Salem
115	Pavithra	AI & DS	1	86	Namakkal

Query 2 - SELECT SPECIFIC COLUMNS

```
SELECT Student_ID, Student_Name, Department, Marks FROM students;
```

Student_ID	Student_Name	Department	Marks
101	Arun	AI & DS	87
102	Divya	CSE	92
103	Kavin	ECE	76
104	Harini	AI & DS	95
105	Priya	IT	81
106	Rahul	CSE	89
107	Keerthana	AI & DS	91
108	Vijay	ECE	73
109	Nandhini	IT	88
110	Sanjay	AI & DS	97
111	Meena	CSE	79
112	Ashwin	IT	84
113	Swetha	ECE	90
114	Dinesh	CSE	94
115	Pavithra	AI & DS	86

Query 3 - WHERE CONDITION

```
SELECT * FROM students WHERE Department = 'AI & DS';
```

Student_ID	Student_Name	Department	Year	Marks	City
101	Arun	AI & DS	1	87	Namakkal

Student_ID	Student_Name	Department	Year	Marks	City
104	Harini	AI & DS	3	95	Karur
107	Keerthana	AI & DS	2	91	Erode
110	Sanjay	AI & DS	4	97	Karur
115	Pavithra	AI & DS	1	86	Namakkal

Query 4 - MARKS ABOVE 90

SELECT Student_Name, Department, Marks FROM students WHERE Marks > 90;

Student_Name	Department	Marks
Divya	CSE	92
Harini	AI & DS	95
Keerthana	AI & DS	91
Sanjay	AI & DS	97
Dinesh	CSE	94

Query 5 - STUDENTS FROM KARUR

SELECT Student_Name, Department, Marks, City FROM students WHERE City = 'Karur';

Student_Name	Department	Marks	City
Harini	AI & DS	95	Karur
Sanjay	AI & DS	97	Karur

Query 6 - MARKS ASCENDING

SELECT Student_Name, Department, Marks FROM students ORDER BY Marks ASC;

Student_Name	Department	Marks
Vijay	ECE	73
Kavin	ECE	76
Meena	CSE	79
Priya	IT	81
Ashwin	IT	84
Pavithra	AI & DS	86
Arun	AI & DS	87
Nandhini	IT	88
Rahul	CSE	89
Swetha	ECE	90
Keerthana	AI & DS	91
Divya	CSE	92
Dinesh	CSE	94
Harini	AI & DS	95
Sanjay	AI & DS	97

Query 7 - MARKS DESCENDING

SELECT Student_Name, Department, Marks FROM students ORDER BY Marks DESC;

Student_Name	Department	Marks
Sanjay	AI & DS	97
Harini	AI & DS	95
Dinesh	CSE	94
Divya	CSE	92
Keerthana	AI & DS	91
Swetha	ECE	90
Rahul	CSE	89
Nandhini	IT	88
Arun	AI & DS	87
Pavithra	AI & DS	86
Ashwin	IT	84
Priya	IT	81
Meena	CSE	79
Kavin	ECE	76
Vijay	ECE	73

Query 8 - AI & DS BY MARKS

SELECT Student_Name, Marks FROM students WHERE Department = 'AI & DS' ORDER BY Marks DESC;

Student_Name	Marks
Sanjay	97
Harini	95
Keerthana	91
Arun	87
Pavithra	86

Query 9 - MARKS BETWEEN 80 AND 90

SELECT Student_Name, Department, Marks FROM students WHERE Marks BETWEEN 80 AND 90 ORDER BY Marks DESC;

Student_Name	Department	Marks
Swetha	ECE	90
Rahul	CSE	89
Nandhini	IT	88
Arun	AI & DS	87
Pavithra	AI & DS	86
Ashwin	IT	84
Priya	IT	81

Query 10 - CSE BY MARKS

SELECT Student_ID, Student_Name, Marks FROM students WHERE Department = 'CSE' ORDER BY Marks DESC;

Student_ID	Student_Name	Marks
114	Dinesh	94
102	Divya	92
106	Rahul	89
111	Meena	79

SQL Concepts Covered

Concept	Purpose
SELECT	Retrieves data from a table.
WHERE	Filters records using a condition.
ORDER BY	Sorts query results.
ASC	Sorts from low to high.
DESC	Sorts from high to low.
BETWEEN	Filters values within a range.

Final Status: TASK 21 COMPLETED SUCCESSFULLY

✓ Database created ✓ 15 records inserted ✓ 10 SQL queries executed ✓ Query outputs documented